Repository Supplement

HardAQG: Automatic Difficult Multiple-Choice Question Generation

This document accompanies the repository for the paper. The repository contains three files:

- `direito.json` — all 50 generation cycles for the Tax Law domain, 45 accepted and 5 discarded, each with attribute scores, quality scores, and failure history.
- `matematica_discreta.json` — all 50 generation cycles for the Discrete Mathematics domain, 44 accepted and 6 discarded, same structure.
- `Relatorio_Prompts_HardAQG.pdf` — the complete text, in Portuguese, of every prompt used by the six agents, in pipeline order, together with a worked example of an accepted question.

This supplement reproduces, in English, two pieces of the prompt content referenced but not fully reproduced in the paper for space, plus the binary conformity criteria checked by the Question Validation Agent (F) alongside the quality scale, mentioned in the paper only by citation to Haladyna et al. It also gives the full references of the two source textbooks and the description of each configurable system parameter, both moved here from the paper for space.

Supplement A. Forbidden Stem Verbs

The Question Generation Agent (D) may not use the following verbs, or the recall-level operation they signal, in a question stem. Six of the seven verbs (list, define, identify, name, recognize, describe) are commonly classified under the Remembering level of Bloom's revised taxonomy, the same level excluded by the cognitive demand attribute (G) at its lowest score. The seventh, summarize, is typically classified one level higher, under Understanding, but is excluded because summarizing a single, explicitly stated passage can still be satisfied by direct restatement, without the synthesis the verb would otherwise imply. As with the other six, the restriction targets the underlying mental operation, not the verb as a lexical item.

Portuguese (original)	English (translation)
listar	list
definir	define
identificar	identify
nomear	name
reconhecer	recognize
descrever	describe
resumir	summarize

Supplement B. Question Validation Agent: Conformity and Quality Criteria

Conformity criteria (binary; any failure rejects the item):

1. *Single correct answer* — exactly one alternative is correct.

2. *No inappropriate content* — no offensive, dangerous, or educationally unsuitable content.
3. *No implausible distractor* — no distractor eliminable without domain knowledge.
4. *No grammatical clue* — no alternative given away by grammatical agreement with the stem.
5. *No absolute terms* — no alternative uses words such as "always", "never", "all", or "none".
6. *No length imbalance* — alternatives have comparable length and specificity.
7. *No combination alternatives* — no alternative of the "all of the above" or "A and C are correct" type.

Quality criteria (three-level ordinal scale):

Criterion	Score 1	Score 2	Score 3
Clarity	Ambiguous: more than one interpretation, a meaning-affecting error, or missing essential information	Clear: single interpretation, grammatically correct, no meaning-affecting ambiguity	Exemplary: maximal precision in stem and alternatives, level-appropriate language, nothing missing
Educational relevance	Irrelevant: tests an administrative or contextual detail rather than content	Partially relevant: touches a real subject, but generically or superficially	Relevant: assesses understanding of a central concept, relation, or application
Distractor quality	Two or more distractors eliminable without domain knowledge	Exactly one distractor eliminable, or none eliminable but not all capture a distinct reasoning error	No distractor eliminable, and each captures a distinct, informative reasoning error
Semantic diversity	Two or more alternatives represent essentially the same idea or conceptual error	Alternatives cover different conceptual positions, but some partially overlap in meaning	Each alternative represents a distinct, mutually exclusive conceptual position

A question passes the Question Validation Agent only if all seven conformity criteria hold and every quality criterion reaches the threshold adopted for the experiment (Table 2 in the main paper).

Supplement C. Source Textbooks

The source material for each domain consists of excerpts from the following textbooks:

- Tax Law: Carneiro C. Curso de Direito Tributário e Financeiro. 9th ed. São Paulo: Saraiva Educação; 2020.
- Discrete Mathematics: Gomide A, Stolfi J. Elementos de Matemática Discreta para Computação. Preliminary version. Universidade Estadual de Campinas; 2011.

Supplement D. System Parameters

Beyond the attribute weights and the acceptance threshold defined in the main paper, the solution exposes the following configurable parameters. The values adopted in the experiment, and the reason for each value, are given in Table 2 of the main paper.

Parameter	Description
Grounding mode	How much the Pedagogical Design Agent's plan may draw on knowledge beyond the retrieved chunks, from strict derivation to bounded external elaboration.
Minimum non-contiguous chunks	Lower bound on non-contiguous chunks the plan must combine, operationalizing locality (L).
Reasoning steps target	Number of steps the plan's reasoning chain must contain, verified by an irreducibility test, operationalizing reasoning depth (R).
Cognitive target (Bloom level)	Level of Bloom's revised taxonomy (reference [1] in the main paper) the plan's required mental operation must reach, operationalizing cognitive demand (G).
Forbidden stem verbs	Verbs the Question Generation Agent may not use in the stem, targeting the mental operation a verb signals rather than the verb itself.
Quality score threshold	Minimum score each quality criterion must reach on the three-level scale applied by the Question Validation Agent.
Maximum attempts per question	Bounds the retry cycle for a question before it is discarded.